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EEE 411 — Past Questions & Materials
by Renance

Course: EEE 411
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 1
Levels: 400

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Paper 1 of 1 — EEE 411 2nd order system

Level 400 • First Semester • Handout

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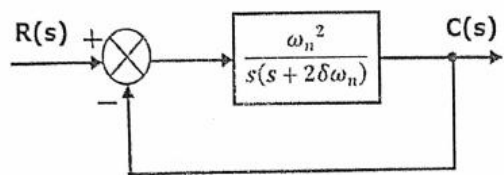
SECOND ORDER SYSTEM

Second order systems exhibit a wide range of responses that must be analysed and described. Varying a first order system parameter simply change the speed of the response whereas changes in the parameters of a second order system can display characteristics much as a first order system depending on the components values, displayed damped and pure oscillation for its transient response.

TIME RESPONSE OF SECOND ORDER SYSTEM

The order of a control system is determined by the power of s in the denominator of its transfer function. If the power of s in the denominator of transfer function of a control system is 2, then the system is said to be second-order control system.

Consider the following block diagram of closed loop control system. Here, an open loop transfer function, $\omega_n^2/s(s+2\delta\omega_n)$ is connected with a unity negative feedback.



The general expression of transfer function of a second order control system is given as:

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1 + G(s)}$$

Substitute, $G(s) = \frac{\omega_n^2}{s(s+2\delta\omega_n)}$ in the equation above

$$\frac{C(s)}{R(s)} = \frac{\left(\frac{\omega_n^2}{s(s+2\delta\omega_n)}\right)}{1 + \left(\frac{\omega_n^2}{s(s+2\delta\omega_n)}\right)} = \frac{\omega_n^2}{s^2 + 2\delta\omega_n s + \omega_n^2}$$

The characteristic equation is

$$s^2 + 2\delta\omega_n s + \omega_n^2 = 0$$

The roots of characteristic equation are

$$s = \frac{-2\delta\omega_n \pm \sqrt{(2\delta\omega_n)^2 - 4\omega_n^2}}{2} = \frac{-2(\delta\omega_n \pm \omega_n\sqrt{\delta^2 - 1})}{2}$$

$$\Rightarrow s = -\delta\omega_n \pm \omega_n\sqrt{\delta^2 - 1}$$

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DYNAMIC BEHAVIOUR OF THE SECOND-ORDER SYSTEM

The dynamic behavior of the second-order system can be described in terms of two parameters δ and ω_n , where, δ and ω_n are damping ratio and natural frequency of the system respectively

- When $\delta = 0$, the two roots are imaginary (Oscillatory – the transient response does not die out)
- When $\delta = 1$, the two roots are real and equal (Critically damped - the transient response does not oscillate)
- When $\delta > 1$, the two roots are real but not equal (Overdamped - the transient response does not oscillate)
- When $0 < \delta < 1$, the two roots are complex conjugate (Underdamped)

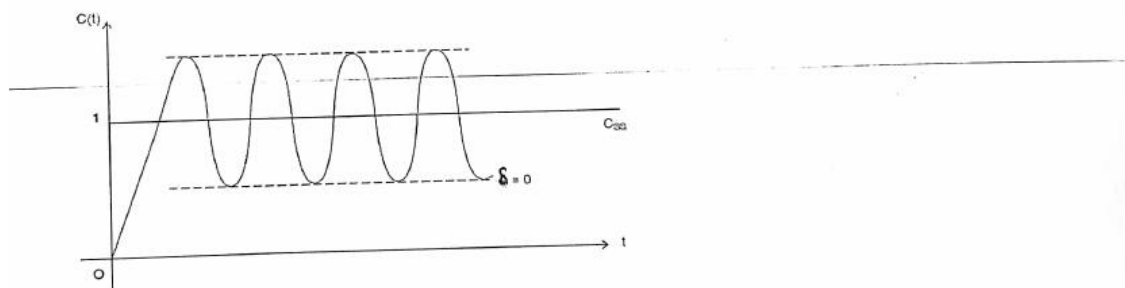
Every practical system takes finite time to reach to its steady state and during this period it oscillates or increases exponentially.

Every system had tendency to oppose the oscillatory behavior of the system which is called damping. This damping is measured by a factor or a ratio called damping ratio of the system. This factor explains, how much dominant the opposition is to the oscillations in the output.

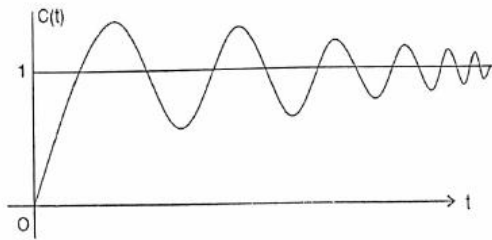
In some system it will be low in which case, system will oscillate but slowly i.e., with damped frequency. Now, this measures the opposition by the system to the oscillatory behavior.

If it is made zero, ($\delta = 0$) system will oscillate with maximum frequency. As there is no opposition from system, system naturally and freely oscillates under such condition. Hence this frequency of oscillation under $\delta = 0$ condition is called Natural frequency of oscillations of the system and denoted by the symbol ω_n (rad/sec).

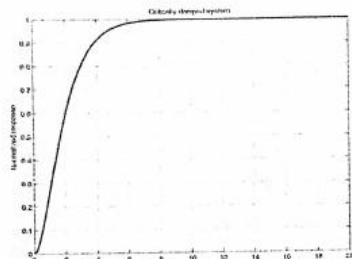
1. Undamped (Sustained Oscillations) $\delta = 0$: A system is said to be sustain damped system when the value of δ is zero. No damping occurs in this case. no opposition condition, system oscillates freely and naturally. Hence this frequency is called as natural frequency.



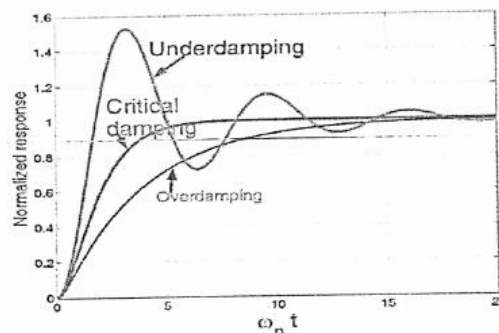
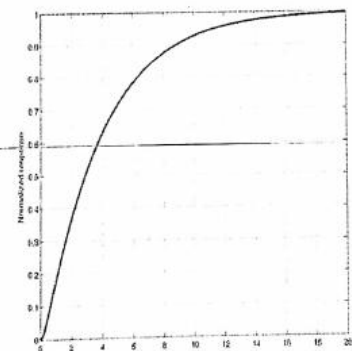
2. Under damped system ($0 < \delta < 1$): A system is said to be under damped system when the value of δ is less than one. In this case roots are complex in nature and the real parts are always negative. System is asymptotically stable. Rise time is lesser than the other system with the presence of finite overshoot.



3. **Critically damped system ($\delta = 1$):** A system is said to be critically damped system when the value of δ is **one**. In this case **roots are real** in nature and the real parts are always repetitive in nature. System is asymptotically stable. **Rise time is less in this system** and there is **no presence of finite overshoot**.



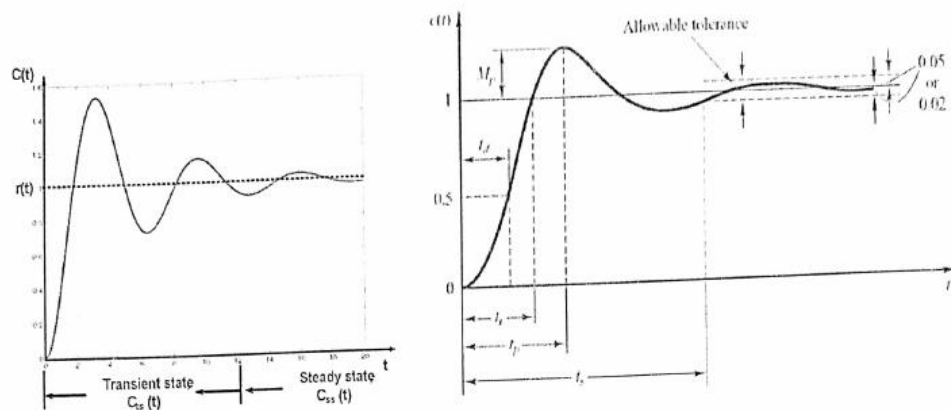
4. **Over damped system ($\delta > 1$):** A system is said to be over damped system when the value of δ is **greater than one**. In this case **roots are real and distinct** in nature and the real parts are always negative. System is asymptotically stable. **Rise time is greater than the other system** and there is **no presence of finite overshoot**.



DEFINITIONS OF TRANSIENT-RESPONSE SPECIFICATIONS.

The performance of the control system can be expressed in the term of transient response to a unit step input function because it is easy to generate. Let us consider a second-order control system in which a unit step input signal is given and it is also considered that the system is initially at rest. That is all initial conditions of the system are zero. The time response characteristics of the system at underdamped condition is drawn below

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All the time domain specifications are represented in above figures. The response up to the settling time is known as transient response and the response after the settling time is known as steady state response.

There are number of common terms in transient response characteristics, which are:

1. **Delay time (t_d)** is the time required to reach 50% of its final value by a time response signal during its first cycle of oscillation. It is the time required for the response to reach **half of its final value** from the zero instant. It is denoted by t_d .

2. **Rise time (t_r)** is the time required to reach final value by an under damped time response signal during its first cycle of oscillation. It is time required for the response to rise from 10% to 90% of the final value for **overdamped** systems and 0 to 100% of the final value for **underdamped** system.

3. **Peak time (t_p)** is simply the time required by response to reach its first peak i.e. the peak of first cycle of oscillation, or first overshoot.

It is the time required for the response to reach the **peak value** for the first time. It is denoted by t_p .

4. **Maximum overshoot (M_p)** or Peak overshoot is defined as the deviation of the response at peak time from the final value of response. It is also called the **maximum overshoot**.

It is the difference between the magnitude of the highest peak of time response and magnitude of its steady state. Maximum overshoot is expressed in term of percentage of steady-state value of the response. As the first peak of response is normally maximum in magnitude, maximum overshoot is simply normalized difference between first peak and steady-state value of a response.

5. **Settling time (t_s)** is the time required for a response to become steady. It is the time required for the response to reach the steady state and stay within the specified tolerance bands around the final value. In general, the tolerance bands are 2% and 5%. The settling time is denoted by t_s . It is time required for the response to reach and stay within a specified tolerance band [usually 2% or 5%] of its final value.

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6. **Steady-state error (ess)** is the difference between actual output and desired output at the infinite range of time. It indicates the error between the actual output and desired output as t tends to infinity and the time taken to reach steady state is called Settling time.

SYSTEM DYNAMICS

System dynamics is the study of characteristic and behaviour of dynamic systems i.e.

- i. Differential equations: First-order systems and Second-order systems,
- ii. Laplace transforms,
- iii. System transfer function,
- iv. Transient response: Unit impulse, Step and Ramp

Laplace transforms convert differential equations into algebraic equations. They are related to frequency response

No.	Function	Time-domain $x(t) = \mathcal{L}^{-1}\{X(s)\}$	Laplace domain $X(s) = \mathcal{L}\{x(t)\}$
1	Delay	$\delta(t-\tau)$	$e^{-s\tau}$
2	Unit impulse	$\delta(t)$	1
3	Unit step	$u(t)$	$\frac{1}{s}$
4	Ramp	t	$\frac{1}{s^2}$
5	Exponential decay	$e^{-\alpha t}$	$\frac{1}{s + \alpha}$
6	Exponential approach	$(1 - e^{-\alpha t})$	$\frac{\alpha}{s(s + \alpha)}$
7	Sine	$\sin \omega t$	$\frac{\omega}{s^2 + \omega^2}$
8	Cosine	$\cos \omega t$	$\frac{s}{s^2 + \omega^2}$
9	Hyperbolic sine	$\sinh \alpha t$	$\frac{\alpha}{s^2 - \alpha^2}$
10	Hyperbolic cosine	$\cosh \alpha t$	$\frac{s}{s^2 - \alpha^2}$
11	Exponentially decaying sine wave	$e^{-\alpha t} \sin \omega t$	$\frac{\omega}{(s + \alpha)^2 + \omega^2}$
12	Exponentially decaying cosine wave	$e^{-\alpha t} \cos \omega t$	$\frac{s + \alpha}{(s + \alpha)^2 + \omega^2}$

STEPS/GUIDELINES FOR SECOND ORDER SYSTEM RESPONSE

We can write $C(s)$ equation as,

$$C(s) = \left(\frac{\omega_n^2}{s^2 + 2\delta\omega_n s + \omega_n^2} \right) R(s)$$

Where,

- $C(s)$ is the Laplace transform of the output signal, $c(t)$
- $R(s)$ is the Laplace transform of the input signal, $r(t)$
- ω_n is the natural frequency
- δ is the damping ratio.

In order to see the response or visualise it, the output $C(s)$ has to be changed back to time domain

Follow these steps to get the response (output) of the second order system in the time domain.

- Take the Laplace transform of the input signal $r(t)$

$$C(s) = \left(\frac{\omega_n^2}{s^2 + 2\delta\omega_n s + \omega_n^2} \right) R(s)$$

- Consider the equation,
- Substitute $R(s)$ value in the above equation
- Do partial fractions of $C(s)$ if required
- Apply inverse Laplace transform to $C(s)$

STEP RESPONSE OF SECOND ORDER SYSTEM

Consider the unit step signal as an input to the second order system.

Laplace transform of the unit step signal is,

$$R(s) = \frac{1}{s}$$

We know the transfer function of the second order closed loop control system is,

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\delta\omega_n s + \omega_n^2}$$

Case 1: $\delta = 0$

Substitute, $\delta = 0$ in the transfer function.

$$\begin{aligned} \frac{C(s)}{R(s)} &= \frac{\omega_n^2}{s^2 + \omega_n^2} \\ \Rightarrow C(s) &= \left(\frac{\omega_n^2}{s^2 + \omega_n^2} \right) R(s) \end{aligned}$$

Substitute, $R(s) = \frac{1}{s}$ in the above equation.

$$C(s) = \left(\frac{\omega_n^2}{s^2 + \omega_n^2} \right) \left(\frac{1}{s} \right) = \frac{\omega_n^2}{s(s^2 + \omega_n^2)}$$

Apply inverse Laplace transform on both the sides.

$$c(t) = (1 - \cos(\omega_n t)) u(t)$$

So, the unit step response of the second order system when $\zeta = 0$ will be a continuous time signal with constant amplitude and frequency.

Case 2: $\zeta = 1$

Substitute, $\zeta = 1$ in the transfer function.

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\omega_n s + \omega_n^2}$$

$$\Rightarrow C(s) = \left(\frac{\omega_n^2}{(s + \omega_n)^2} \right) R(s)$$

Substitute, $R(s) = \frac{1}{s}$ in the above equation.

$$C(s) = \left(\frac{\omega_n^2}{(s + \omega_n)^2} \right) \left(\frac{1}{s} \right) = \frac{\omega_n^2}{s(s + \omega_n)^2}$$

Do partial fractions of $C(s)$.

$$C(s) = \frac{\omega_n^2}{s(s + \omega_n)^2} = \frac{A}{s} + \frac{B}{s + \omega_n} + \frac{C}{(s + \omega_n)^2}$$

After simplifying, you will get the values of A, B and C as 1, -1 and $-\omega_n$ respectively. Substitute these values in the above partial fraction expansion of $C(s)$.

$$C(s) = \frac{1}{s} - \frac{1}{s + \omega_n} - \frac{\omega_n}{(s + \omega_n)^2}$$

Apply inverse Laplace transform on both the sides.

$$c(t) = (1 - e^{-\omega_n t} - \omega_n t e^{-\omega_n t}) u(t)$$

So, the unit step response of the second order system will try to reach the step input in steady state.

Case 3: $0 < \delta < 1$

We can modify the denominator term of the transfer function as follows –

$$\begin{aligned} s^2 + 2\delta\omega_n s + \omega_n^2 &= \{s^2 + 2(s)(\delta\omega_n) + (\delta\omega_n)^2\} + \omega_n^2 - (\delta\omega_n)^2 \\ &= (s + \delta\omega_n)^2 + \omega_n^2(1 - \delta^2) \end{aligned}$$

The transfer function becomes,

$$\begin{aligned} \frac{C(s)}{R(s)} &= \frac{\omega_n^2}{(s + \delta\omega_n)^2 + \omega_n^2(1 - \delta^2)} \\ \Rightarrow C(s) &= \left(\frac{\omega_n^2}{(s + \delta\omega_n)^2 + \omega_n^2(1 - \delta^2)} \right) R(s) \end{aligned}$$

Substitute, $R(s) = \frac{1}{s}$ in the above equation.

$$C(s) = \left(\frac{\omega_n^2}{(s + \delta\omega_n)^2 + \omega_n^2(1 - \delta^2)} \right) \left(\frac{1}{s} \right) = \frac{\omega_n^2}{s((s + \delta\omega_n)^2 + \omega_n^2(1 - \delta^2))}$$

Do partial fractions of $C(s)$.

$$C(s) = \frac{\omega_n^2}{s((s + \delta\omega_n)^2 + \omega_n^2(1 - \delta^2))} = \frac{A}{s} + \frac{Bs + C}{(s + \delta\omega_n)^2 + \omega_n^2(1 - \delta^2)}$$

After simplifying, you will get the values of A, B and C as 1, -1 and $-2\delta\omega_n$ respectively. Substitute these values in the above partial fraction expansion of $C(s)$.

$$\begin{aligned} C(s) &= \frac{1}{s} - \frac{s + 2\delta\omega_n}{(s + \delta\omega_n)^2 + \omega_n^2(1 - \delta^2)} \\ C(s) &= \frac{1}{s} - \frac{s + \delta\omega_n}{(s + \delta\omega_n)^2 + \omega_n^2(1 - \delta^2)} - \frac{\delta\omega_n}{(s + \delta\omega_n)^2 + \omega_n^2(1 - \delta^2)} \\ C(s) &= \frac{1}{s} - \frac{(s + \delta\omega_n)}{(s + \delta\omega_n)^2 + (\omega_n\sqrt{1 - \delta^2})^2} - \frac{\delta}{\sqrt{1 - \delta^2}} \left(\frac{\omega_n\sqrt{1 - \delta^2}}{(s + \delta\omega_n)^2 + (\omega_n\sqrt{1 - \delta^2})^2} \right) \end{aligned}$$

Substitute, $\omega_n\sqrt{1 - \delta^2}$ as ω_d in the above equation.

$$C(s) = \frac{1}{s} - \frac{(s + \delta\omega_n)}{(s + \delta\omega_n)^2 + \omega_d^2} - \frac{\delta}{\sqrt{1 - \delta^2}} \left(\frac{\omega_d}{(s + \delta\omega_n)^2 + \omega_d^2} \right)$$

Apply inverse Laplace transform on both the sides.

$$c(t) = \left(1 - e^{-\delta\omega_n t} \cos(\omega_d t) - \frac{\delta}{\sqrt{1 - \delta^2}} e^{-\delta\omega_n t} \sin(\omega_d t) \right) u(t)$$

$$c(t) = \left(1 - \frac{e^{-\delta\omega_n t}}{\sqrt{1 - \delta^2}} \left((\sqrt{1 - \delta^2}) \cos(\omega_d t) + \delta \sin(\omega_d t) \right) \right) u(t)$$

If $\sqrt{1 - \delta^2} = \sin(\theta)$, then ' δ ' will be $\cos(\theta)$. Substitute these values in the above equation.

$$c(t) = \left(1 - \frac{e^{-\delta\omega_n t}}{\sqrt{1 - \delta^2}} (\sin(\theta) \cos(\omega_d t) + \cos(\theta) \sin(\omega_d t)) \right) u(t)$$

$$\Rightarrow c(t) = \left(1 - \left(\frac{e^{-\delta\omega_n t}}{\sqrt{1 - \delta^2}} \right) \sin(\omega_d t + \theta) \right) u(t)$$

So, the unit step response of the second order system is having damped oscillations (decreasing amplitude) when ' δ ' lies between zero and one.

Case 4: $\delta > 1$

We can modify the denominator term of the transfer function as follows:

$$\begin{aligned} s^2 + 2\delta\omega_n s + \omega_n^2 &= \{s^2 + 2(s)(\delta\omega_n) + (\delta\omega_n)^2\} + \omega_n^2 - (\delta\omega_n)^2 \\ &= (s + \delta\omega_n)^2 - \omega_n^2 (\delta^2 - 1) \end{aligned}$$

The transfer function becomes,

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{(s + \delta\omega_n)^2 - \omega_n^2 (\delta^2 - 1)}$$

$$\Rightarrow C(s) = \left(\frac{\omega_n^2}{(s + \delta\omega_n)^2 - \omega_n^2 (\delta^2 - 1)} \right) R(s)$$

Substitute, $R(s) = \frac{1}{s}$ in the above equation.

$$C(s) = \left(\frac{\omega_n^2}{(s + \delta\omega_n)^2 - (\omega_n \sqrt{\delta^2 - 1})^2} \right) \left(\frac{1}{s} \right) = \frac{\omega_n^2}{s(s + \delta\omega_n + \omega_n \sqrt{\delta^2 - 1})(s + \delta\omega_n - \omega_n \sqrt{\delta^2 - 1})}$$

Do partial fractions of $C(s)$.

$$C(s) = \frac{\omega_n^2}{s(s + \delta\omega_n + \omega_n\sqrt{\delta^2 - 1})(s + \delta\omega_n - \omega_n\sqrt{\delta^2 - 1})}$$

$$= \frac{A}{s} + \frac{B}{s + \delta\omega_n + \omega_n\sqrt{\delta^2 - 1}} + \frac{C}{s + \delta\omega_n - \omega_n\sqrt{\delta^2 - 1}}$$

After simplifying, you will get the values of A, B and C as 1 , $\frac{1}{2(\delta + \sqrt{\delta^2 - 1})(\sqrt{\delta^2 - 1})}$ and $\frac{-1}{2(\delta - \sqrt{\delta^2 - 1})(\sqrt{\delta^2 - 1})}$ respectively. Substitute these values in above partial fraction expansion of $C(s)$.

$$C(s) = \frac{1}{s} + \frac{1}{2(\delta + \sqrt{\delta^2 - 1})(\sqrt{\delta^2 - 1})} \left(\frac{1}{s + \delta\omega_n + \omega_n\sqrt{\delta^2 - 1}} \right)$$

$$- \left(\frac{1}{2(\delta - \sqrt{\delta^2 - 1})(\sqrt{\delta^2 - 1})} \right) \left(\frac{1}{s + \delta\omega_n - \omega_n\sqrt{\delta^2 - 1}} \right)$$

Apply inverse Laplace transform on both the sides.

$$c(t) = \left(1 + \left(\frac{1}{2(\delta + \sqrt{\delta^2 - 1})(\sqrt{\delta^2 - 1})} \right) e^{-(\delta\omega_n + \omega_n\sqrt{\delta^2 - 1})t} \right.$$

$$\left. - \left(\frac{1}{2(\delta - \sqrt{\delta^2 - 1})(\sqrt{\delta^2 - 1})} \right) e^{-(\delta\omega_n - \omega_n\sqrt{\delta^2 - 1})t} \right) u(t)$$

Since it is over damped, the unit step response of the second order system when $\delta > 1$ will never reach step input in the steady state.